

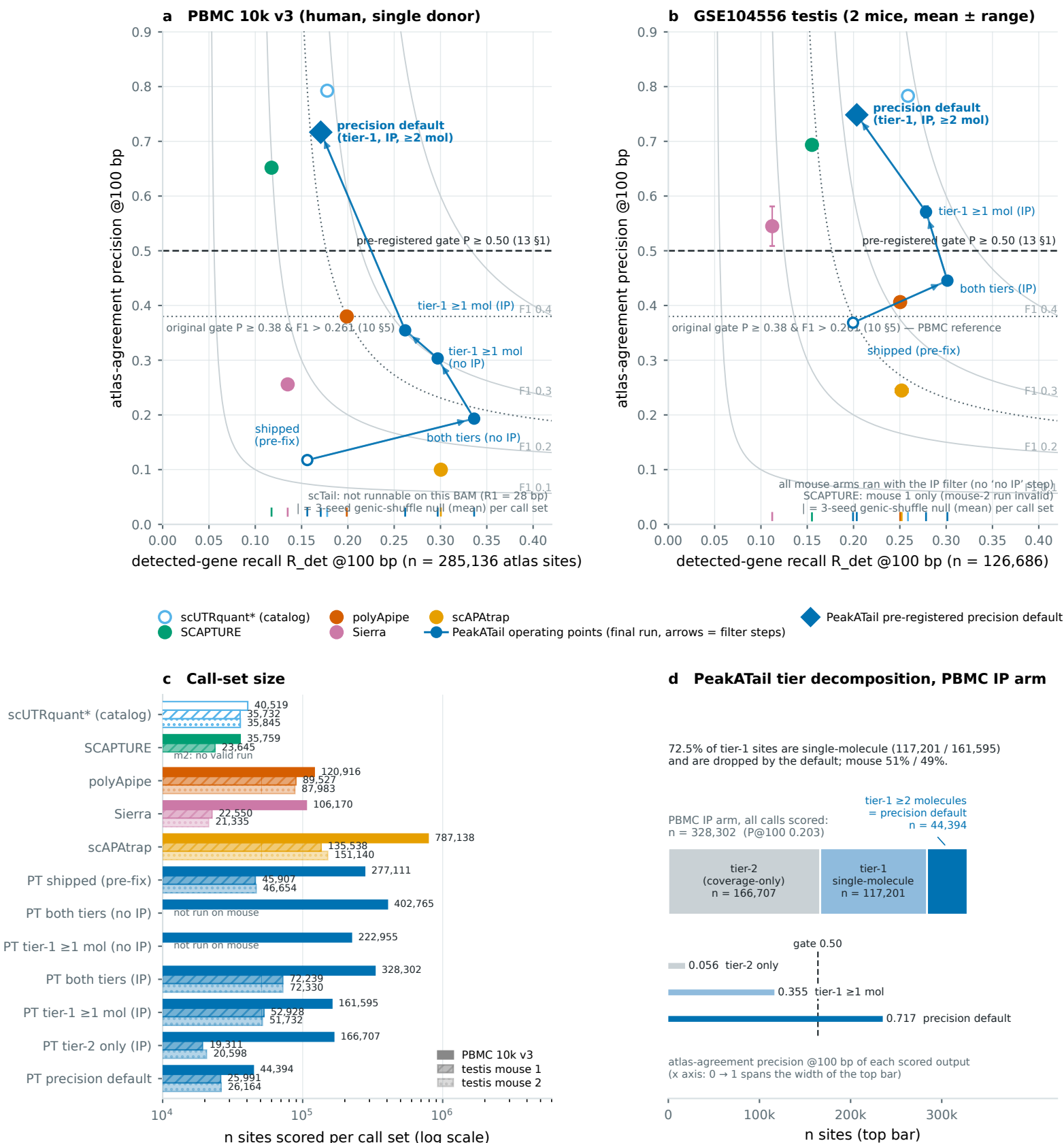


Fig 2 | Trustworthy PAS detection: the final Stage-2 run (PeakATail code 4efeb125) against the competitor panel. Definitions: atlas-agreement precision = fraction of called sites with a strand-matched PolyASite 2.0 representative site within 100 bp (point distance); it is agreement with a curated atlas, not ground truth (atlas-novel true sites count as false positives). Recall is detected-gene recall R_{det} : denominator = atlas sites in genes detected in the dataset (285,136 human / 126,686 mouse). Full-atlas recall of the precision default is 0.086 (PBMC; 569,005 sites) and 0.096 / 0.098 (mice; 301,006 sites). One scorer (score_tool.py), same denominators for every tool; 3-seed gene-body-shuffled nulls. scUTRquant is catalog-based (hollow; shown, not ranked). Pre-registration: the precision-first default and its gate ($P@100 \geq 0.50$ on PBMC and both mice) were committed before the run (gate committed 01:19, arms started 02:59 (2026-08-21); 13 \$1, 15); the ≥ 2 -molecule threshold was chosen after a stale Stage-2 sweep and pre-registered before this run (12 CORRECTION). Gate: PASS on all three call sets (0.717 / 0.741 / 0.755); no ≥ 1 -molecule output clears the original $P \geq 0.38$ floor. Caveats: single PBMC donor / single Cell Ranger BAM; the two mice are one study and chemistry; the default's recall is at or below polyApipe's (-14% PBMC, -19% mouse) and the $F1_{det}$ lead (+0.015 / +0.010) is not a meaningful margin — the claim is precision, not overall accuracy. The non-IP PBMC ≥ 2 -molecule file (pas_tier1_ge2mol_noIP_POSTHOC) is not the pre-registered default and is not shown. Every plotted value: results/figures/manuscript/final_benchmark*.tsv.